

Aria Kidr

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EDUCATION

University of Birmingham

Sep. 2024 – June 2028

MSci Theoretical Physics

Achieved **First Class Honours** in First and Second Year

Sandwell Academy

Sep. 2017 – June 2024

A-Level: Mathematics (A*), Physics (A) and Further Mathematics (A).

GCSE: **1x Grade 8, 4x Grade 7, 3x Grade 6, 1x Grade 5** including **Mathematics (8)** and **Physics (7)**

WORK EXPERIENCE

University of Birmingham

June 2026 – Aug. 2026

Summer Research Intern

Sun, Stars, and Exoplanets Group

- Created an autoencoder fit for denoising Stellar Spectra data of power and frequency.
- Produced artificial data as a learning curve to observe and tweak the behaviour of the neural network.
- Proceeded to utilise real data from the Kepler mission using thousands of red giant stars.
- Documented everything I have done as a separate file available on GitHub.

University of Birmingham

June 2025 – Sep. 2025

Summer Research Assistant

Theoretical Physics Group

- Conducted research under two professors on computational modelling of phase transitions.
- Developed a Python program to simulate the 2D Ising Model, implementing Hamiltonian energy and magnetisation calculations.
- Applied the Metropolis Algorithm to model thermal fluctuations and system evolution.
- Documented code with step-by-step comments to improve clarity and reproducibility.

MyTutor

Oct. 2024 – Present

Tutor

Remote

- Currently tutoring Mathematics and Physics at A-Level and GCSE.
- Tutoring students online, enabling them to develop problem-solving and independent learning skills for academic and personal success.
- Developed strong communication skills through explaining technical material clearly and concisely in an online setting.
- Gained experience in time management and self-organisation while balancing tutoring with university studies.

DH Furniture Group Ltd

July 2022 – Sep. 2022

Furniture Assembler

Birmingham, Perry Barr

- Operated and maintained a CNC (Computer Numerical Control) machine for precision cutting.
- Assembled furniture pieces following technical specifications and quality standards.
- Developed practical skills in handling machinery, troubleshooting minor issues, and ensuring workplace safety.
- Worked effectively as part of a team to meet production targets and deliver high-quality finished products.

SKILLS AND AWARDS

- **Awards:** The Birmingham Project award for collaborating in a multidisciplinary team (Siemens 3) to address the challenge on how Siemens technologies be integrated to support sustainable education on campus.
- **Programming Languages:** Python, LaTeX, Bash, HTML
- **Skills:** Teamwork, Time Management, Problem Solving, Critical Thinking, Organised and Adaptable.
- **Languages:** English, Kurdish (Sorani).
- **Interests:** Swimming, Badminton and Writing Movie Reviews.